Anil Sahith Vallepu

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Professional Summary

Software Engineer with strong Math and CS foundations, specializing in AI/ML systems and production-grade deployment. Proven expertise in developing and optimizing deep learning models, LLM applications, and RAG systems for real-world use cases across healthcare, finance, and enterprise workflows. Skilled in building low-latency inference pipelines, on-device model optimization, and scalable data architectures.
Seeking roles: AI Engineer, Machine Learning Engineer, LLM Engineer, Data Scientist, Quant ML Developer.

Skills

- **Programming Languages**: Python, C++, JavaScript, TypeScript, SQL, MATLAB.
- **Tools & Platforms**: VS Code, PyCharm, Jupyter Notebook, Cursor, Figma, Postman, GitHub, Vercel.
- **Web Technologies**: React.js, Next.js, Node.js, Flask, FastAPI, RESTful APIs, WebSocket, Streamlit, Gradio.
- **Databases & Cloud**: MySQL, PostgreSQL, MongoDB, Redis, Pinecone, FAISS, ChromaDB, AWS, GCP, Docker.
- **Machine Learning & AI**: TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, OpenCV, NLP, RAG, Knowledge Graphs, LLMs, Hugging Face Transformers, Langchain.
- **Other Coursework**: Data Structures and Algorithms, System Design, Embedded Systems, Edge Computing, LLM Fine-Tuning and Inference Optimisation, Quantum Computing and PQC, Quant Finance.

Experience

- **Noccarc Robotics Pvt Ltd** — June 2024 - April 2025
  AI Engineer — *Pune, Maharashtra*
  - Architected a real-time arrhythmia detection system for embedded patient monitoring devices using deep learning, achieving 95%+ accuracy across 18+ cardiac arrhythmia classes on the PhysioNet database.
  - Implemented a hybrid neural network combining ResNet, CBAM attention mechanism, and GRU layers with separable convolutions for efficient edge deployment, optimised inference pipeline to achieve ∼300ms latency, processing ECG data every 5 seconds while consuming under 12% single-core CPU resources.
  - Established an end-to-end data pipeline from ECG sensor integration to frontend visualisation and backend processing with automating alarm generation, created 20+ distinct alarm classifications for critical heart abnormalities, reducing clinical response times by 40% in critical care environments.
- **Carelon Global Solutions (Elevance Healthcare)** — June 2023 - August 2023
  Summer Intern - AI/ML — *Bangalore, Karnataka*
  - Automated document processing pipeline using computer vision and deep learning to extract and classify data from 700+ documents per hour with 95% accuracy; deployed an end-to-end OCR solution using PyTesseract for text extraction, OpenCV for image preprocessing, and YOLOv8 object detection for region identification.
  - Accelerated processing performance through multi-threading and multi-processing techniques, reducing document processing time by 90% and decreasing operational costs, delivered a production-ready solution by integrating with existing data workflows for scalable enterprise document management.

Projects

- **MiniAI-As-A-Service App**: Designed a full-stack web application for automated CSV data analysis along with machine learning model training and AI-generated insights; implemented FastAPI backend for model training, prediction, and statistical analysis endpoints with a Redis caching layer and PostgreSQL for persistent storage.[**Link**]
- **SlackBot Content Creation AI Assistant**: Launched an AI-powered Slack bot streamlining content research, keyword analysis, and competitive analysis workflows; incorporated multiple APIs for web scraping, search engine results, and email notifications via SendGrid, reducing content planning time from 6+ hours to 15 minutes for marketing teams. [**Link**]
- **HSN Code Classification System**: Constructed an intelligent chatbot system using RAG and Knowledge Graphs for accurate HSN code classification; leveraged NetworkX to model hierarchical HSN code relationships and deployed a user-friendly web interface using Streamlit with 92% classification accuracy. [**Link**]
- **Dynamic AI Exam-Prep App**: Developed an adaptive learning platform with a multi-agent orchestration system for personalised NCLEX exam preparation; programmed an intelligent scheduling algorithm for dynamic study calendar generation based on performance metrics and integrated a contextual AI chatbot tutor. [**Link**]

Education

- **National Institute of Technology (NIT) Warangal** — Warangal, India
  Bachelor of Technology in Electrical and Electronics Engineering. — *May 2024*

Achievements

- Secured 99.3rd percentile in JEE Mains (India) - Top 0.7% among 1 million+ candidates
- Recognised with Top 100 Projects at NASA AMES Space Settlement Contest, San Juan, Puerto Rico (2017)
- Solved 300+ Data Structures and Algorithms problems (Medium-Hard difficulty) on LeetCode